

The Psychology of Brain Rot

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Abstract

Brain rot refers to the cognitive decline which occurs primarily from the consumption of content which is considered to be trivial and unchallenging, particularly online content. Brain rot can be a result of social media addiction or stimulating oneself with content that can cause deviations in normal cognitive patterns. This is a descriptive research paper which aims to obtain an understanding of the psychology behind brain rot and ways in which it can be dealt with, such as dopamine detox. Although the topic is relatively new and requires further research, simple psychological interventions are potentially effective in dealing against brain rot, since it is not an issue of clinical significance as of now.

Keywords: Brain rot, social media, cognitive deterioration.

Introduction

Operational Definition

Brain rot, defined by Oxford University (2024), is “the supposed deterioration of a person’s mental or intellectual state, especially viewed as the result of overconsumption of material (now particularly online content) considered to be trivial or unchallenging. Also: something characterized as likely to lead to such deterioration.”

Brief History

The word “brain rot” first appeared in a book authored by Henry David Thoreau (1854) titled “Walden”. The book focuses on simple living, geographically set in a wood cabin, near a pond called “Walden Pond.” The term “brain rot” was used to describe the effects of one’s lifestyle on their intellect. The word was rarely used, but slowly gained popularity as the rise of social media occurred in the digital age. The word was popularly used in social media applications such as “Discord.” Nowadays, content that classifies as brain rot, runs rampant on most public social media platforms such as Instagram. In 2024, Oxford University Press declared it as “the word of the year.”

Basic Understanding of Brain Rot

In the current era, brain rot mainly occurs from online social media consumption, particularly from content that is trivial and may lead to one's cognitive deterioration. It follows a simple cycle of consuming social media content, experiencing a sense of fatigue and consuming more social media content, which in the long term, leads to a decline in cognitive abilities. The content in general, also is a determinant in the efficiency by which it can decline an individual's cognitive ability. Content that causes "sensory over-stimulation" such as pornography, videos with rapidly changing images or online content which can be considered relative aversive, is axiomatically regarded to lead to cognitive overload, which may cause psychological fatigue in an individual. Therefore, such content can be classified as "Brain rot inducing." The term "brain rot" is also used to refer to the content which causes it, and not just the content itself. For example, sexually inappropriate and aversive videos can be called "brain rot" directly, rather than calling them "brain rot inducing."

Review of Literature

Neurological Basis of Cognitive Deterioration through Addiction

Cognitive deterioration refers to the decline in the ability to perform cognitive functions such as decision-making, working memory, problem-solving, etc. One noticeable motivator behind consuming online content is the need for stimulation and reward. The neurologically, the receptor responsible for the anticipation of reward and consequent stimulation is the D1 dopaminergic receptor (Witt et al. 2025). The continuous activation and phosphorylation of this receptor results in a protein called "arrestin" binding to the receptor, which reduces its activity (Kim et al. 2004). Therefore, we can infer that constant stimulation by social media would lead to the desensitisation of dopamine receptors. Consequently, dopamine desensitisation would lead to a progressive loss of motivation to perform tasks which do not provide the same degree of stimulation (Wise 2004) along with the development of addiction to the stimulus (Worley 2007), in this case, social media. This would potentially decrease the motive to engage in cognitive tasks which would aid in cognitive development. Engagement in complex cognitive tasks and learning, in order to adapt to our environment is mediated by the process of synaptic plasticity, which results from synchronised neuronal firing patterns (McAllister & Lo). A lack of motivation to engage in learning, would lead to a decline in synaptic plasticity in neuronal firing patterns for complex cognitive tasks, which is a key neurological factor for developing cognitive function. Hence, we can infer that constantly stimulating ourselves with social media, would lead to a reduction in synaptic plasticity relevant to complex cognitive tasks and would result in cognitive deterioration. This describes the state of brain rot from a "addiction-motivation" perspective.

Cognitive deviation in humour

Cognitive deviation usually refers to a change in typical cognitive abilities. The ability to process and appreciate humour is a complex cognitive ability which has many psychological

explanations. The three main theories of human often discussed in psychology are the incongruence theory, the relief theory, and the superiority theory. As discussed by de Silva (2018) in the “theories of humour” chapter of the book “The Psychology of Humour and Emotions in Buddhism”, incongruence was one of the central theories to his study and it created “cracks and fissures” between two contradicting emotions, leading to humour (de Silva 2018). Content relevant to brain rot, if humorous, mainly follows incongruity and relief theory of humour, since people consume brain rot inducing content for entertainment and relief and the content is usually trivial or deviant from how a normal individual would project their thoughts on the topic of discussion shown in the content. For example, dark humour is generally considered to be brain rot inducing, and the content of dark humour involves comedically depicting what would normally be considered tragic or controversial, hence making it a form of incongruent humour. Although incongruent humour is generally not a form of cognitive deviation, since it was broadly used historically, there are limits beyond which it may lead to cognitive deviation. A study conducted by Voisey & Heintz (2024) involved a sample of (N = 160) participants. The data was collected through an online survey which assessed dark humour, the dark tetrad and online trolling. With respect to dark humour, the results of the study had shown that the traits of the dark tetrad (Psychopathy, Narcissism, Machiavellianism, and Sadism) were linked to the ability to enjoy as well as produce dark humour (Voisey & Heintz 2024). A study by Papousek et al. (2017) used EEG coherence to check the prefrontal-posterior coupling with respect to humour. Loose prefrontal-posterior coupling showed loosening of the control of the prefrontal cortex and was associated with more malicious humour styles (Papousek et al. 2017). Brain rot inducing content usually entails dark humour, hence we can infer that it may lead to reduced prefrontal control and development of dark tetrad traits, which would evidently be a form of cognitive deviation if it grows extreme.

Potential Solutions to Brain Rot

According to Niwlikar (2024), some ways of dealing with brain rot include dopamine detox, physical movement, sleep, social connection, and mindfulness. These methods resemble the biopsychosocial model of health proposed by Engel (1977). According to this model, the health of an individual has three determinants: biological factors, psychological factors, and social factors. Although brain rot is psychological, psychological factors of health can be affected physical health (Ohrnberger et al. 2017) and social health (Kawachi & Berkman 2001). From this we can infer that the negative effects of brain rot can be dealt with by improving one’s social and physical well-being, along with their psychological well-being. Since motivation is primarily required to advance with one’s goal towards improving well-being and motivation being mediated by dopamine, it is necessary to perform a dopamine detox in order to have oneself motivated to improve their physical and social health. Dopamine detox would involve reducing or cutting off social media use until the brain’s reward circuits have been repaired to the extent where the individual would feel pleasure from cognitive tasks which are less stimulating in comparison to social media consumption. Dopamine detox and mindfulness aid with psychological health and healthy exercise, diet, and sleep can aid with physical health. Social health can be improved by acquiring social capital. Social capital, a term used by Bayraktar (2025) used to define “the networks of

relationships among people who live and work in a particular society.” It can affect mental health through social support, sense of belonging, and access to resources (Bayraktar 2025).

Discussion

From our overall understanding, we know that brain rot, although used previously in a historical context, has gained popular usage in the current era, particularly in the context of social media usage. An important concern regarding the psychology of brain rot includes the impact of how social media may deteriorate cognitive function of future generations. As social media usage has gained its popularity, the negative effects of social media have also been elucidated from the observed effects on future generations. Social media addiction has grown prominent, with a global prevalence of 18.4% and with a prevalence of 22.8% in Asia, among university students (Salari et al. 2025). A study by Cheng et al. (2021) showed that pooled prevalence estimate for social media addiction was 31% in collectivistic nations and 14% in individualistic nations. From this we can infer that there is a high risk of individuals experience brain rot, since it is commonly a result of social media addiction. With high addiction rates in the current generation, there may be a decline in motivation to complete tasks which may be of cognitive challenge, hence leading to cognitive decline and consequently, brain rot. As discussed in the review of literature, this can be dealt with by performing a dopamine detox, which would enable individuals to potentially escape their social media addiction before integrating themselves with a healthier lifestyle.

Psychoeducation is another recommended measure to prevent the future generation from engaging in social media usage patterns (i.e.- overuse and consumption of psychologically harmful digital content) and consequently experiencing brain rot. Another related terms of discussion, which are also associated with brain rot, is “doom-scrolling.” This is a repetitive behaviour which occurs from the need for stimulation (as previously discussed under the role of addiction) usually occurs in social media platforms such as Instagram or YouTube, which provides short videographic content for brief viewing, usually navigated through “scrolling” through the application. Due to the short duration of the content, the individual consuming it would want to consume more to meet their need for stimulation, hence they will continuously scroll until it becomes a compulsive habit. An implication for this would be a reduction in one’s attention span. Since scrolling through content, exposes an individual to constantly changing stimuli, each distinct from the other, it may hypothetically promote the individual to pay less attention to the longitudinal course of any content which doesn’t have the same rate of change. A study by Yan et al. (2024) used an attention network test and recorded the data through electroencephalography (EEG) on a sample of 48 participants. The results of the study showed that “an increased tendency toward mobile phone short video addiction could negatively impact self-control and diminish executive control within the realm of attentional functions.” (Yan et al. 2024) From this we can infer that doom scrolling would affect one’s attention span, which is a cognitive function, hence doom-scrolling can potentially be classified as a determinant for brain rot. Overall, we can say that social media platforms should be used wisely and their harmful consequences should be educated to the general public, especially the future generation.

Conclusion

Brain rot is generally caused by social media addiction or consuming content that may influence an individual to deviate from normal cognitive patterns. Potentially, brain rot can be reversed by performing a dopamine detox and convincing oneself to live a healthier lifestyle psychologically, physically, and socially. Brain rot may be a future threat for the next generation, since social media consumption is growing rampant and therefore psychological interventions such as psychoeducation must be implemented in order to teach the next generation about the risks of social media and how it should be used without having to deal with its negative consequences.

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